WQD7009 Big Data Applications and Analytics

Mid Term Assessment

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Date: 7th December 2025

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1 Introduction to the Google Cloud Data Analytics Certificate Path

Google Skills is a learning online platform by Google that provides a comprehensive list of training and certifications related to Google Cloud Platform (GCP) skills and services in different domains. These domains include AI/ML, Security, Infrastructure, DevOps, Data and many more.

The selected certificate path is called *Beginner: Google Cloud Data Analytics Certificate* and would fall under the Data domain, and it consists of 5 enriching courses. The 5 courses and their contents are as below:

1. Introduction to Data Analytics in Google Cloud
   It introduces core concepts of cloud-based data analytics, explains the role of a data analyst, and provides an overview of GCP's data analytics services and tools.

2. Data Management and Storage in the Cloud
   This course focuses on how to effectively store and manage data leveraging various storage options (Cloud Storage, BigQuery, Cloud SQL, Bigtable), as well as data organization strategies, data lifecycle management etc.

3. Data Transformation in the Cloud
   This course teaches data processing and transformation techniques using GCP tools, covering data cleaning, preprocessing, ETL/ELT processes, and working with tools like Dataflow, Dataprep, and BigQuery for data transformation.

4. The Power of Storytelling: How to Visualize Data in the Cloud
   This course emphasizes data visualization and communication skills. It covers creating effective dashboards and reports using tools like Looker Studio, while emphasizing on storytelling.

5. Put It All Together: Prepare for a Cloud Data Analyst Job
   The final course that integrates all previous learnings. It covers end-to-end analytics workflows, portfolio development, interview preparation, and real-world scenarios that data analysts encounter.

2 Lessons Learned

2.1 Contents Covered

There are a total of 4 modules covered as follow:

Module	Content
Module 1: Introduction to Cloud Computing	1. Cloud computing infrastructure and service models 2. History of cloud computing 3. Benefits of cloud computing 4. BigQuery Introduction
Module 2: Data Analytics in the Cloud	1. Cloud migration 2. Cloud deployment models 3. Cloud's role in data analytics and benefits 4. Costs optimization 5. The Google Cloud Architecture Framework
Module 3: The Data Lifecycle	1. Importance of Data Management 2. Cloud Safety and Privacy 3. Data Lifecycle 4. Roles on Data Team 5. Automation in Data Lifecycle 6. Data Retention Policies 7. Version Controls and Holds
Module 4: The Role of Cloud Data Analyst	1. Google's Cloud Data Analytics Tools 2. End-to-end Data Management 3. Data Access Management 4. Business Data Request 5. Dataproc Introduction 6. Process Management

2.2 Interesting and Challenging Highlights

There are many contents that I find interesting and meaningful, taking firstly an example of the lab for navigating **BigQuery**. I appreciate the practicality of the lab, being able to carry out different big data related operations on the environment that reflect the actual UI and dataset. It first starts with adding the NBA dataset, then querying the relevant information, subsequently extending it to Looker to visualise the queried data. This mutli-phase step-by-step learning guide provides a comprehensive understanding of the purpose of different services on Google Cloud Platform and how to leverage them in a chained and systematic approach. This hands-on aspect of this course allows me to obtain heightened understanding beyond theoretical knowledge which is crucial for real-world application during work later.

Next, as someone that values high-level framework and strategy, I find the **Google Cloud Architecture Framework** particularly exciting. The framework consists of 6 pillars: (1) System design as the foundation, (2) Operational excellence, (3) Security, privacy, and compliance, (4) Reliability, (5) Cost optimisation and (6) Performance optimisation. I personally believe that it is

incredibly crucial to have guiding principles as such to allow anyone working with the cloud to have a set of guidelines as reference for better alignment. In particular, the third pillar regarding security, privacy and compliance is an aspect that many might not realise the importance of as the immediate effect is not very visible compared to others. Hence, I value that data security, privacy and compliance is highlighted as an important pillar since sensitive data that contains PII (personally identifiable information) such as name, email address, home address are addressed and masked accordingly since it is a fundamental right for users to have their information properly safeguarded.

There were certain challenging learning points, and the one that I would like to elaborate on is the activity on **designing the data lifecycle plan** roleplaying as a data analyst at TheLook eCommerce. Data lifecycle plan can be a complex subject as it covers 6 stages, from the initial plan stage to the last destroy stage. Each phase has to be thoroughly understood its essence and meticulously planned out in order for the data lifecycle to be effective. The activity was challenging as it requires reading and understanding of information from varying sources and subsequently synthesising everything to produce a business slide. Primarily, this activity highlights the importance of a well-planned data lifecycle plan as well and providing a glimpse to the challenges it might bring.

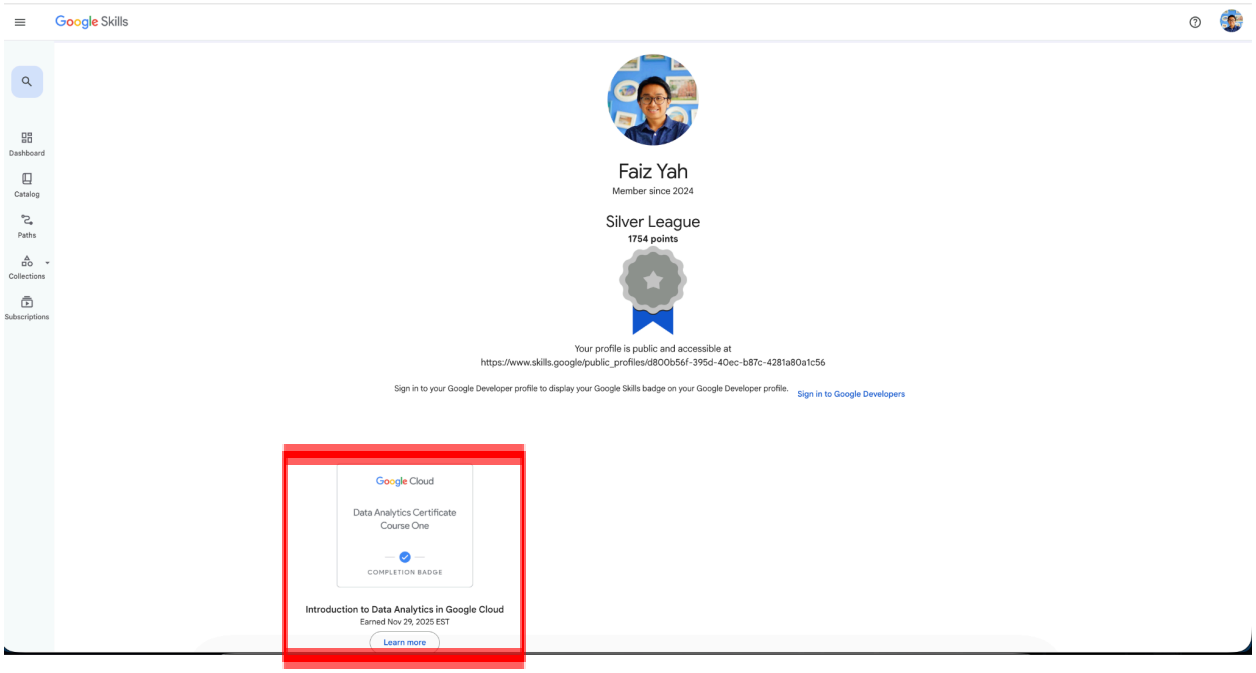
2.3 Contributions Towards Academic and Career

Reflecting and looking back at the contents that were introduced, I believe it contributes considerably for both academic and career. As big data is becoming more and more common, added on the fact that machine learning performance depends significantly on the completeness of the big data, this implies that the content taught here will all be highly relevant. Taking an example of the **BigQuery lab and Dataproc** introduction, I now will be able to leverage this service for my future projects, storing my data, querying them for EDA and create my models end-to-end fully on the cloud.

From a career standpoint, I personally believe it also adds value. Speaking from personal experience already working in the industry, although different companies might use varying platforms (AWS, GCP, Azure), the fundamental concepts all remain the same, which the course has introduced well. For instance, it explains the **job scope and responsibility of different data job titles,** including Data Analyst, Data Engineer, Data Scientist and Data Architect, allowing students to comprehend in detail and choosing a direction that best suits them.

Appendix

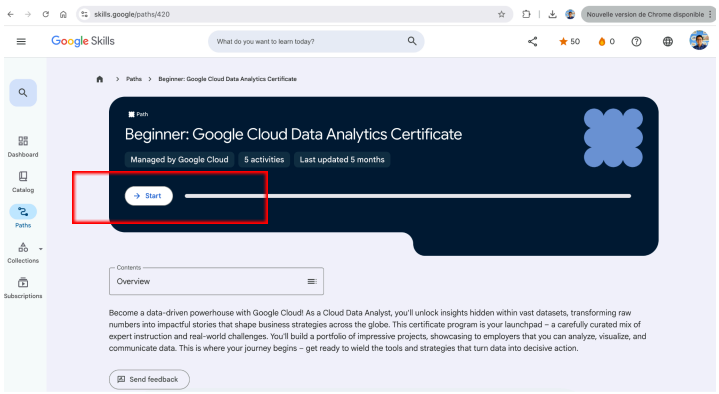
i) Public Profile with Badge Completion

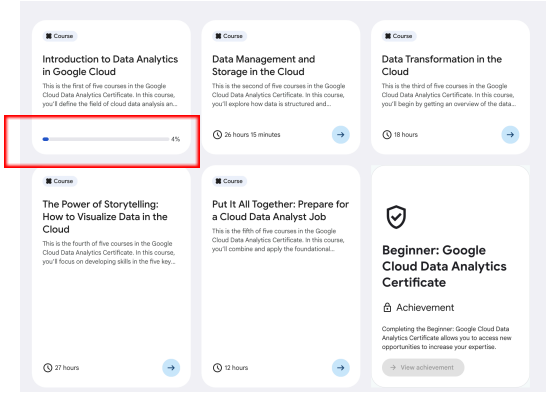


Screenshot of Profile (link)

ii) Before Completion of Course

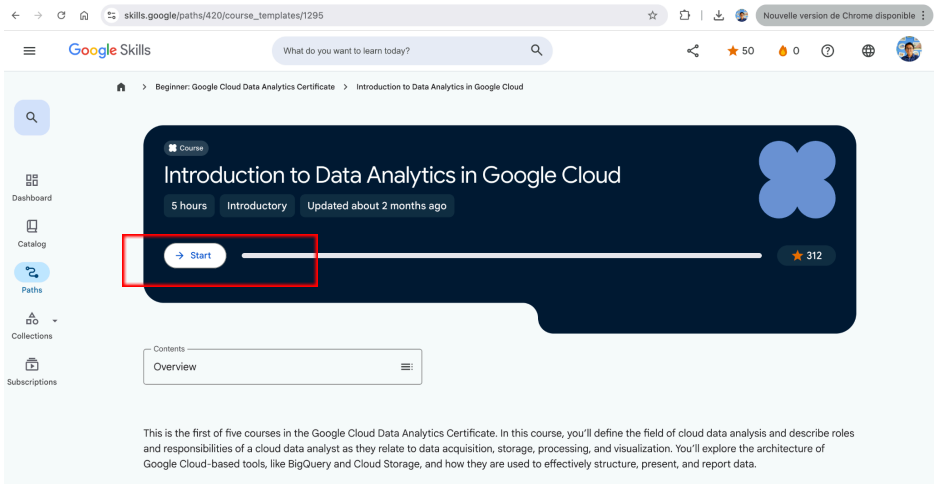
a. Certificate Path





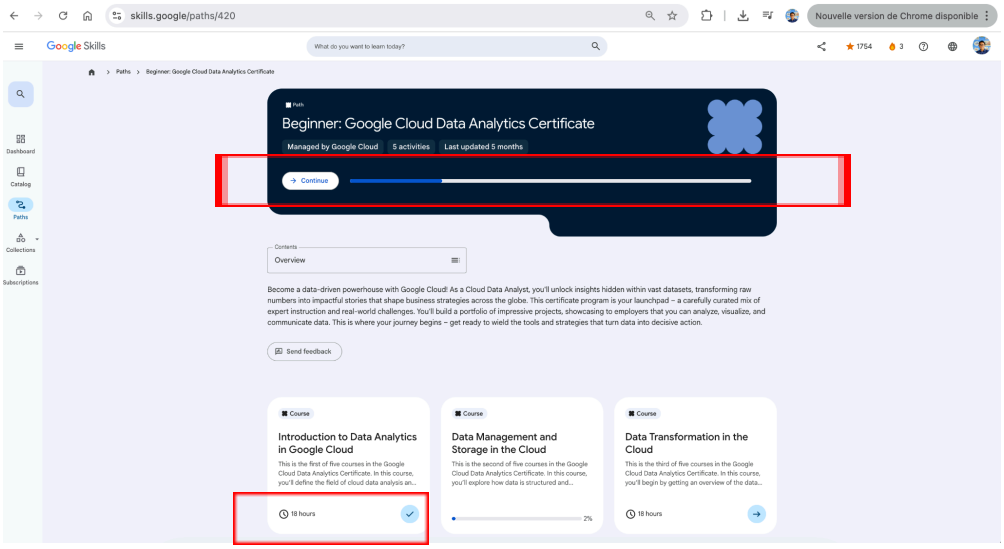
Screenshot of Certificate Path (Before Completion of Course)

b. Course



Screenshot of Course (Before Completion of Course)

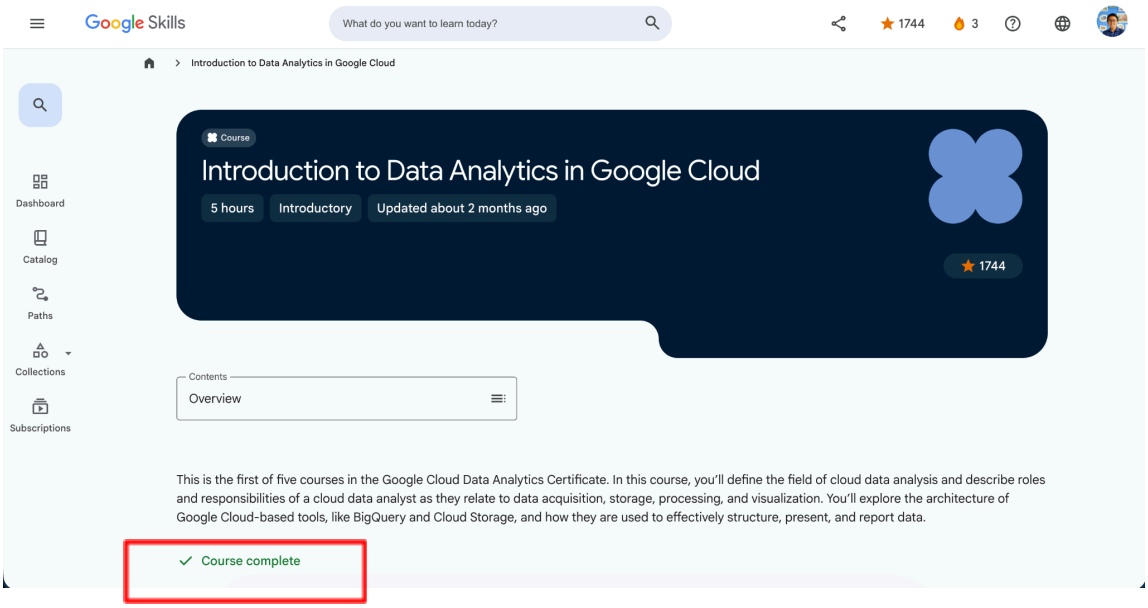
iii) After Completion of Course



a. Certificate Path

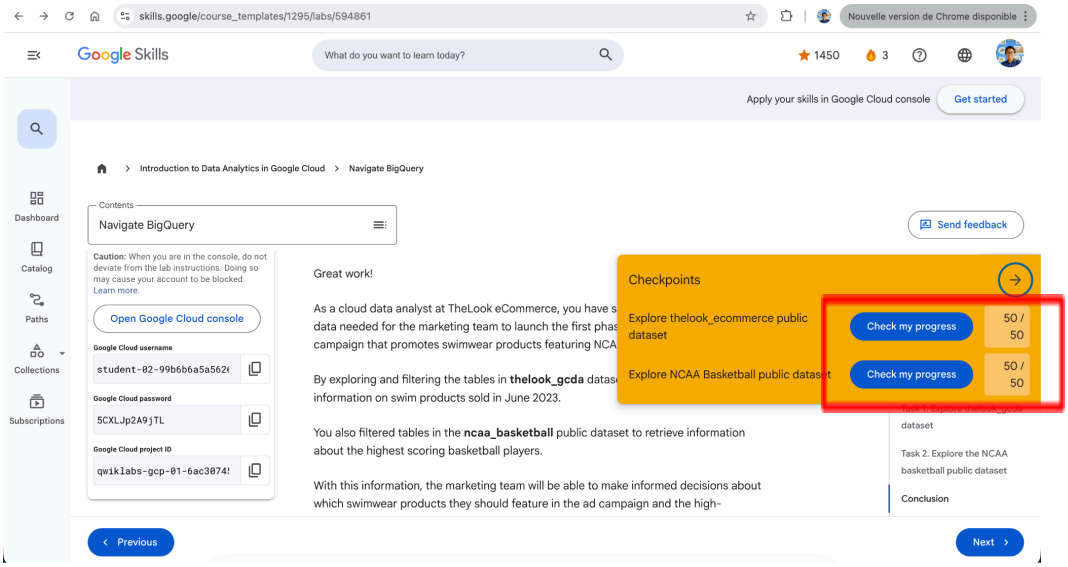
Screenshot of Certificate Path (After Completion of Course)

b. Course



Screenshot of Course (After Completion of Course)

c. Lab (BigQuery)



Screenshot of Completed Lab (BigQuery)